

Challenge (Habanero)

Q1. Simplify the expression $\sqrt{64x^3}$.

- (A) $8x\sqrt{x}$
- (B) $4x\sqrt{x}$
- (C) $8x\sqrt{3}$
- (D) $4\sqrt{x}$
- (E) $8\sqrt{x}$

Q2. The data usage of a gaming console is measured as $\sqrt{2t^2 + 5t}$, where t is the time in hours. Simplify the expression.

- (A) $\sqrt{2t}(t + 2.5)$
- (B) $\sqrt{t}(2t + 5)$
- (C) $t\sqrt{2t + 5}$
- (D) $\sqrt{2t^2 + 5}$
- (E) $\sqrt{t}(2t + 2.5)$

Q3. The score of a gamer in a competition is given by $\sqrt[3]{s^2}$, where s is the number of points. Simplify the expression.

- (A) $s^{\frac{2}{3}}$
- (B) $s^{\frac{1}{3}}$
- (C) s^2
- (D) $s^{\frac{4}{3}}$
- (E) $s^{\frac{3}{2}}$

Q4. The coding logic for a game involves the expression $\sqrt{3p^2 - 2p + 1}$. Simplify the expression.

- (A) $\sqrt{3p^2 - 2p + 1}$
- (B) $p\sqrt{3 - \frac{2}{p} + \frac{1}{p^2}}$
- (C) $\sqrt{3p} - 1$
- (D) $p\sqrt{3 - 2 + 1}$
- (E) $\sqrt{p^2}$

Q5. The data storage of a gaming laptop is measured as $\sqrt[3]{d^2}$,

where d is the storage capacity. Convert the expression to rational exponent form.

- (A) $d^{\frac{1}{3}}$
- (B) $d^{\frac{2}{3}}$
- (C) $d^{\frac{3}{2}}$
- (D) $d^{\frac{2}{5}}$
- (E) $d^{\frac{4}{3}}$

Q6. The logic of a game involves the expression $\sqrt{x^2 + 2x + 1}$. Simplify the expression.

- (A) $x^2 + 2x + 1$
- (B) $x + 1$
- (C) $x^2 - 2x + 1$
- (D) $x^2 + x$
- (E) $x - 1$

Q7. The score of a gamer in a tournament is given by $\sqrt[3]{g^2 + 2g + 1}$, where g is the number of points. Convert the expression to rational exponent form.

- (A) $g^{\frac{2}{3}} + 2g^{\frac{1}{3}} + 1$
- (B) $g^{\frac{2}{3}} + g^{\frac{1}{3}}$
- (C) $g^{\frac{1}{3}} + g^{\frac{2}{3}}$
- (D) $g^{\frac{1}{3}}$
- (E) $g^{\frac{2}{3}}$

Q8. The coding logic of a game involves the expression $\sqrt{x^2 - 4x + 4}$. Simplify the expression.

- (A) $x - 2$
- (B) $x + 2$
- (C) $x^2 - 4x + 4$
- (D) $x^2 + 4x$
- (E) $2x$

Open Ended Questions (FRQ)

Q9. A gaming console has a data storage capacity of d terabytes. If the data is stored in a $\sqrt{d}$ terabyte folder, and the folder is further divided into $\sqrt[3]{d}$ smaller folders, determine the expression for the total number of smaller folders.

Q10. The scoreboard of a gaming tournament displays the top 5 scores. If the scores are s_1, s_2, s_3, s_4 , and s_5 , and the expression $\sqrt{s_1^2 + s_2^2 + s_3^2 + s_4^2 + s_5^2}$ represents the total score, simplify the expression and determine the value of s_1 if the total score is 100.

Chef's Tips

- Use concise language and provide step-by-step explanations
- Check for simplification and rationalization of radicals
- Emphasize the importance of accuracy and precision